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United States Patent Application Publication

20250283620

Kind Code

A1

Publication Date

September 11, 2025

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VARIABLE GEOMETRY MANIFOLD FOR GENERATING LAMINAR AIR FLOW

Abstract

Systems, methods, and apparatus for generating a laminar air flow using a variable geometry manifold.

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Family ID: 96948908

Appl. No.: 18/597854

Filed: March 06, 2024

Publication Classification

Int. Cl.: F24F8/192 (20210101); F24F8/30 (20210101); F24F13/02 (20060101)

U.S. Cl.:

CPC F24F8/192 (20210101); F24F8/30 (20210101); F24F13/02 (20130101);

Background/Summary

TECHNICAL FIELD

[0001] The present invention generally relates to the generation of directed laminar air flow, and in particular systems, apparatus, and methods for generating laminar air flow through electrostatic precipitator air purifier devices for removing particulate matter.

BACKGROUND

[0002] Outdoor pollution is one of the primary environmental threats to human health, especially in

urban environments. Despite the known adverse health effects, there does not exist a simple and cost-effective solution for wide-area airborne particulate matter pollution. Attempts to address outdoor air purification have been limited due to the nature of available technologies and their application at scale. For example, traditional indoor air purification systems generally use filter-based systems to capture particulate matter from a controlled air-flow. However, such filter-based systems are expensive, and require continual replacement of expensive filters such that it is not feasible for removing large amounts of particulate pollution from outdoor environments.

[0003] To overcome these disadvantage, the use of electrostatic precipitators (ESP) has been proposed. For example, as proposed by Allen et al., in U.S. application Ser. No. 18/204,930, modified ESP air purifier devices can be configured to operate at scale in outdoor environments. As described by Allen, outdoor air containing neutrally charged airborne particulate matter is introduced into an ESP air purifier. The particulate matter is ionized by an electric field generated between positively and negatively charged plates. The ionized particulate matter, now carrying a negative charge, is repelled by the negatively charged plate to the positively charged plate where it can be captured for further processing and eventual removal, while the purified air flow is reintroduced to the external environment. Despite this advancement in the field, the sheer volume of particular matter in outdoor environments requires ESP devices to process large volumes of external air in an efficient and effective manner. As such, there exists a need to increase the efficiency of particulate matter collection in ESP devices operating in outdoor environments.

SUMMARY OF THE INVENTION

[0004] In one aspect, the invention includes a variable geometry manifold for generating a laminar air flow through an ESP device. In this aspect, a variable geometry manifold includes an axially symmetric intake having a proximal and terminal end, wherein the terminal end is narrower than the proximal end, and wherein the axially symmetric intake is configured to be in fluid communication with an electrostatic precipitator (ESP) air purifier. A suction fan mounted to the terminal end of the axially symmetric intake can be configured to draw external air containing particulate matter where it passes from the proximal end to the terminal end of the axially symmetric intake and is converted to a laminar flow of air prior to being introduced to the ESP air purifier.

[0005] In another aspect, the present invention includes a system for electrostatically purifying air using a variable geometry manifold. In this preferred aspect, an axially symmetric intake having a proximal and terminal end, wherein the terminal end is narrower than the proximal end, and is further configured to be in fluid communication with an electrostatic precipitator (ESP) air purifier. A suction fan can be mounted to the terminal end of the axially symmetric intake and configured to draw external air containing particulate matter that is neutrally charged through the axially symmetric intake. In this aspect, the axially symmetric intake forms a laminar flow of the external air that is introduced to the ESP air purifier where it passes through an electric field that extends from a negatively charged plate to a positively charged collecting plate. The electric field ionizes the particulate matter within the laminal flow, such that the electric field repels the ionized particulate matter away from the negative plate and towards the collecting plate causing the ionized particulate matter to attach to the collecting plate.

[0006] In another aspect, external air, preferably containing particulate matter can be passed through the axially symmetric intake, for example by the natural flow of air generated by a wind force, or by having the axially symmetric intake secured to a moving object or device such that external air is directed through the intake as a result of its movement in space.

[0007] In another aspect, the axially symmetric intake includes a non-continuous axially symmetric intake.

[0008] In another aspect, the axially symmetric intake includes a continuous axially symmetric intake.

[0009] In a preferred aspect, the continuous axially symmetric intake includes a plurality of intake

panels coupled by one or more in-line brackets.

[0010] In another preferred aspect, the continuous axially symmetric intake includes at least four intake panels coupled by a plurality of in-line brackets.

[0011] In a preferred aspect, the plurality of in-line brackets externally couple the lateral surfaces of the four intake panels forming a joint that maintains a continuous inner surface of the axially symmetric intake.

[0012] In another aspect, the variable geometry manifold further includes an intake assembly coupled to the proximal end of the axially symmetric intake.

[0013] In a preferred aspect, the intake assembly includes a plurality of assembly panels coupled through one or more panel attachments.

[0014] In a preferred aspect, the intake assembly includes a coupler rotatably securing at least one assembly panel.

[0015] In a preferred aspect, the intake assembly includes an intake gate, which can preferably include a slatted gate or mesh screen.

[0016] In another aspect, the variable geometry manifold further includes a support base securing the axially symmetric intake in-line with the air flow into the body of an ESP air purifier.

[0017] In a preferred aspect, the support base includes a plurality of support arms secured to the proximal end of the axially symmetric intake.

[0018] In a preferred aspect, the support base includes one or more support columns secured to the terminal end of the axially symmetric intake.

[0019] In another aspect, the variable geometry manifold further includes mounting plate configured to secure the suction fan to the one or more support columns, wherein the suction fan is secured in-line with the air flow into the body of an ESP air purifier.

[0020] Additional aspects of the inventive technology will become apparent from the specification, figures and claims below.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0021] FIG. 1A-D: (A) shows a front perspective view of a variable geometry manifold coupled with an intake assembly and support base in one embodiment thereof; (B) shows a top view of a variable geometry manifold coupled with an intake assembly in one embodiment thereof; (C) shows a side view of a variable geometry manifold and support base coupled with an intake assembly and support base in one embodiment thereof; and (D) shows a bottom perspective view of a variable geometry manifold coupled with an intake assembly in one embodiment thereof.

[0022] FIG. 2: shows a front perspective view of a variable geometry manifold coupled with an intake assembly and support base in one embodiment thereof.

[0023] FIG. 3A-E: (A) shows a front perspective view of an isolated axially symmetric intake in one embodiment thereof; (B) shows a front perspective view of an isolated intake assembly in one embodiment thereof; (C) shows a front perspective view of an isolated support base in one embodiment thereof; (D) shows a front perspective view of an isolated intake gate in one embodiment thereof; and (E) shows a front perspective view of a variable geometry manifold in one embodiment thereof.

[0024] FIG. 4A-C: (A) shows a blow-up image of an in-line bracket securing the longitudinal edges of two intake panels in one embodiment thereof; (B) shows a front perspective view of an isolated axially symmetric intake in one embodiment thereof; and (C) shows a top view of an isolated axially symmetric intake in one embodiment thereof.

[0025] FIG. 5A-B: (A) shows a front perspective view of an isolated support base in one embodiment thereof; and (B) shows a rear view of an isolated support base in one embodiment

thereof.

[0026] FIG. 6A-B: (A) shows a side view of an isolated support base coupled with an axially symmetric intake in one embodiment thereof; and (B) shows a front perspective view of an isolated support base coupled with an axially symmetric intake in one embodiment thereof.

[0027] FIG. 7: shows a front perspective view of an intake assembly in one embodiment thereof.

[0028] FIG. 8: show a support base coupled with an intake assembly and a suction fan positioned adjacent to the anticipated terminal end of the axially symmetric intake in one embodiment thereof.

[0029] FIG. 9: shows a schematic diagram of an exemplary ESP air purifier in one embodiment thereof.

DETAILED DESCRIPTION OF THE INVENTION

[0030] The present invention includes a variety of aspects, which may be combined in different ways. The following descriptions are provided to list elements and describe some of the embodiments of the present invention. These elements are listed with initial embodiments, however, it should be understood that they may be combined in any manner and in any number to create additional embodiments. The variously described examples and preferred embodiments should not be construed to limit the present invention to only the explicitly described systems, techniques, and applications. Further, this description should be understood to support and encompass descriptions and claims of all the various embodiments, systems, techniques, methods, devices, and applications with any number of the disclosed elements, with each element alone, and also with any and all various permutations and combinations of all elements in this or any subsequent application.

[0031] The present invention includes systems, methods, and apparatus for the high-efficiency removal of particulate matter from outdoor environments using ESP air purifiers. As described herein, the efficiency of particulate ionization and capture from outdoor spaces can be improved through the generation of a consistent laminar flow of external air through the ESP device and preferably an ESP air purifier as described by Allen et al., in U.S. application Ser. No. 18/204,930, (the ESP device described therein, and its methods of use being incorporated herein by reference).

[0032] As generally shown in FIG. 9, ESP air purifiers can be utilized to remove airborne particulate matter (PM) from air. A diagram illustrating a perspective view of an ESP air purifier in accordance with an embodiment of the invention is shown in FIG. 9. The ESP air purifier (100) includes a negative plate (106) and one or more collecting plates such as, but not limited to, a first collecting plate (104) and a second collecting plate (114). In many embodiments, the negative plate (106) may become negatively charged, as described further below. In several embodiments, the negative plate (106) may take various shapes and sizes. For example, in some embodiments, the negative plate (106) may be a flat, metal sheet. In many embodiments, the first collecting plate (104) and the second collecting plate (114) may become positively charged, as described further below. In several embodiments, the at least one collecting plate (e.g., the first collecting plate (104) and the second collecting plate (114)) may take various shapes and sizes. For example, the first and second collecting plates (104), (114) may be flat, metal sheets.

[0033] In reference to FIG. 9, the negative plate (106) may be positioned in between the first collecting plate (104) and the second collecting plate (114). In some embodiments, the first and second collecting plates (104), (114) may be in parallel. In some embodiments, the negative plate (106) may be positioned parallel to the first collecting plate (104) and the second collecting plate (114). The negative plate (106) and the at least one collecting plate (104), (114) may be spaced depending on various factors such as, but not limited to, the desired strength of an electric field to be generated, input voltage, etc., as further described below. For example, in some embodiments, the first collecting plate (104) and the second collecting plate (114) may be positioned within 10 cm of the negative plate.

[0034] In further reference to FIG. 9, PM in the air (120) may be pulled into the ESP air purifier and become negatively charged (i.e., “ionized”) when passing through an electric field between the

negative plate (106) and the first and/or second collecting plate (104), (114), as described further below. In many embodiments, the ESP air purifier includes a suction fan (116) configured to pull air (120) into the ESP air purifier (100). In some embodiments, the suction fan (116) is a slow-speed fan. In several embodiments, the ESP purifier (100) may also include an exhaust fan (110) configured to release purified air (112) out of the ESP air purifier (100). In some embodiments, the ESP air purifier (100) may also include a nozzle and/or one or more baffles, as described further below.

[0035] In further reference to FIG. 9, the ESP air purifier (100) may include one or more rappers such as, but not limited to, a first rapper (102) configured to vibrate and dislodge the ionized PM from the first collecting plate (104) and a second rapper (118) configured to vibrate and dislodge the ionized PM from the second collecting plate (114). In many embodiments, the first rapper (102) may not be in direct contact with the first collecting plate (104). For example, the first rapper (102) may be connected to the first collecting plate (104) through a first support 103. In some embodiments, the first support 103 may be made of insulator material. In several embodiments, the first rapper (102) may be active to vibrate the first support 103, which in turn may vibrate the first collecting plate (104) in dislodging the ionized PM from the first collecting plate (104), as discussed further below. In many embodiments, the second rapper (118) may not be in direct contact with the second collecting plate (114). For example, the second rapper (118) may be connected to the second collecting plate (114) through a second support (122). In some embodiments, the second support (122) may be made of insulator material. In several embodiments, the second rapper (118) may be activated to vibrate the second support (122), which in turn may vibrate the second collecting plate (114) in dislodging the ionized PM from the second collecting plate (114), as discussed further below.

[0036] In further reference to FIG. 9, the ESP air purifier (100) may include a collection tray (124) may be positioned below the at least one collecting plate (e.g., first collecting plate (104) and the second collecting plate (114)). In many embodiments, the collection tray (124) may collect the dislodged ionized PM from the first collecting plate (104) and/or the second collecting plate (114). In some embodiments, the collection tray (124) may be similar or the same size and dimensions as a bottom of the ESP air purifier (100).

[0037] The ESP air purifier (100) with the collection tray (124) removed from the ESP air purifier (100) in accordance with an embodiment of the invention is shown in FIG. 1B. As described above, the first and second rappers (102), (118) (via the first and second supports 103, (118)) may vibrate and dislodge the ionized PM from the first and second collecting plates (104), (114), respectively, and collected on the collection tray (124). In several embodiments, the collection tray (124) may be accessed and cleaned to remove the ionized PM accumulated in the collection tray (124). In various embodiments, the collection tray (124) configured to be removed from the ESP air purifier (100) using various methods and/or devices that would be known to one of ordinary skill in the art. For example, in some embodiments, the collection tray (124) may slide out of the ESP air purifier (100). In some embodiments, the collection tray (124) may be held to the ESP air purifier (100) using one or more hinges where the collection tray (124) may swing open for cleaning. In some embodiments, the collection tray (124) may include one or more layers. For example, a top layer of the collection tray (124) may be configured to collect the dislodged ionized PM and the top layer may be removed for cleaning/replacement.

[0038] The present invention includes a variable geometry manifold (200) configured to be in fluid communication with an ESP air purifier (100) configured to remove particulate matter, preferably from an external environment. As used herein, the variable geometry manifold (200) of the invention includes an axially symmetric intake (202) having an opening at its proximal end (B) to permit transit of external air, and preferably outdoor air, into the manifold, and an intermediate portion adapted to have a geometric configuration that changes size across all three-dimensions, and a terminal end (A) that is narrower than the opening. In one preferred embodiment, the axially

symmetric intake (202) includes a funnel-like apparatus that changes size across all three-dimensions. Notably, the axially symmetric intake (202) changing size across all three-dimensions can be continuous, or non-continuous in certain embodiments.

[0039] The variable geometry manifold (200) includes an axially symmetric intake (202) having a terminal and proximal end (A, B). Generally referring to the preferred embodiment shown in FIG. 4, the terminal end includes an external aperture (208) configured to be positioned in fluid communication with an external environment. As further shown in FIG. 4, the external aperture (208) of the axially symmetric intake (202) decreases along an intermediate portion across all three-dimensions to form an internal aperture (210) at the terminal end of the device. In this configuration, external air passing through the external aperture (208) is directed in a controlled manner through the intermediate portion of the axially symmetric intake (202) toward the smaller internal aperture (210).

[0040] By directing the external air flow (212) through a narrowing channel having a continually reduced cross-section, the axially symmetric intake (202) causes the disordered external air to increase in speed and be converted into a laminar air flow (212). As used herein, “laminar air flow” refers to a process where air is manipulated to force air to move at the same speed and in the same direction, with no, or minimal cross-over of air streams in a given space. Similarly, “laminar air flow design” may refer to, for example, a variable geometry manifold that contains one or more asymmetrical intakes that are designed to generate laminar air flow through an electrostatic precipitator with no or minimal cross-over of air streams in a given space.

[0041] The axially symmetric intake (202) of the invention can include an integral component, such as a unitary cast or molded component. In alternative embodiments, the axially symmetric intake (202) of the invention can be formed from a plurality of disparate components. In one embodiment, a plurality of intake panels (204) can be coupled together by one or more in-line brackets (206). As used herein, the terms “coaxial,” “in-line,” and “coaxially aligned,” means two or more components generally oriented or positioned as having a common axis.

[0042] As shown in FIG. 4c, four intake panels (204) forming a continuous axially symmetrical shape can be secured lengthwise along their longitudinal edges by one more in-line brackets (206). As highlighted in FIG. 4c, in this preferred configuration, the in-line brackets (206) can be configured to be positioned flush with the top surface of the intake panels (204) forming a joint that allows for the internal surface of the intake panels (204) to form a continuous surface.

[0043] As further shown in FIG. 4, the proximal end (B) of the axially symmetric intake (202), being narrower than the terminal end (A), forms an internal aperture (210) that is in fluid communication with an electrostatic precipitator (ESP) air purifier device (100) as described herein. In this embodiment, the internal aperture (210) is configured to facilitate the directed laminar air flow containing particular matter into the body of the ESP for processing and collection. As used herein, an “electrostatic precipitator,” also referred to as an “ESP,” or “ESP air purifier” includes the systems, methods and apparatus shown in U.S. application Ser. No. 18/204,930 for an electrostatic precipitator (ESP) air purifier configured to remove particulate matter from an air flow, the specific designs and structures of which are incorporated herein by reference in their entirety.

[0044] The current invention includes an intake assembly (214) configured to be coupled with a proximal surface of the axially symmetric intake (202). The intake assembly (214) of the invention can include a single unitary integral component, such as a cast or injection model component. As shown in FIG. 7, in a preferred embodiment, the intake assembly (214) of the invention can be formed by a plurality of interconnected assembly panels (216). In this embodiment, each assembly panel (216) includes an integral panel attachment (220) configured to be secured to an adjacent assembly panel (216) forming the body of intake assembly (214).

[0045] In another embodiment, one or more assembly panels (216) can be rotatably coupled with the intake assembly (214). As shown in FIG. 7, in this embodiment an assembly panel (216) is

rotatably secured with a coupler (216) allowing it to be rotated outward allowing access to the internal compartment of the intake assembly (214). The rotatable assembly panel (216) can be positioned over one or more gaskets (218) secured to the adjacent free edge of the corresponding assembly panels (216) forming a seal.

[0046] As further shown in FIG. 7, an intake gate (222) can be secured to the front surface of the intake assembly (214) through one or more coupling devices (not shown). In this preferred embodiment, the intake gate (222) can swing open and allow access to the internal compartment of the intake assembly (214). The intake gate (222) of the invention can further secure a screen, such a metal mesh screen, or slatted gate to prevent large objects, such as trash or other similar items from entering the intake assembly (214) and being transmitted through the axially symmetric intake (202).

[0047] The present invention further includes a support base (224). Referring to FIG. 6, the support base (224) of the invention can be configured to position the axially symmetric intake (202) and intake assembly (214) approximately in-line with the internal compartment of the ESP device (100). In this configuration, the support base (224) aligns the air flow entering the axially symmetric intake (202) and facilitates the generated laminar air flow directly into the ESP device (100) without interruption. In the preferred embodiment shown in FIG. 6, the support base (224) of the invention can include a pair of support arms (226) that can be secured to the proximal end of the axially symmetric intake (202). The support base (224) of the invention can further include one or more support columns secured to the terminal end of the axially symmetric intake (202), the height of which can be adjusted to align with the internal compartment of ESP device (100).

[0048] Finally, as shown in FIG. 8, a suction fan (230) can be secured to the terminal end (A) of the axially symmetric intake (202). In a preferred embodiment, a mounting plate (232) can be secured to the external surface of the terminal end (A) of the axially symmetric intake (202) thereby positioning the suction fan (230) flush with the internal aperture (210).

[0049] In other embodiment, external air can be passed through the axially symmetric intake (202) without the use of a suction fan (230) as described above. For example, in one example a variable geometry manifold (200) of the invention can be secured to a stationary object (not shown), such a building or other external structure, and positioned such that a naturally, or non-naturally occurring wind force can cause external air to pass through the axially symmetric intake (202) and converted into a laminar air flow (212) as generally described herein. In another preferred example, a variable geometry manifold (200) of the invention can be secured to a device or other object that can independently move causing external air to pass through the variable geometry manifold (200). For example, a variable geometry manifold (200) of the invention can be secured to a non-stationary object (not shown), such as vehicle, train, subway, or boat and further positioned such that the axially symmetric intake (202) is directed into the flow of air cause by the movement of the underlying vehicle thereby causing the disordered external air to be converted into a laminar air flow (212) as generally described herein. In this manner, one or more variable geometry manifold (200) can coupled with an ESP device and used to purify the internal environment of a vehicle, such as a car, bus, or ship while in motion.

[0050] Naturally, all embodiments discussed herein are merely illustrative and should not be construed to limit the scope of the inventive technology consistent with the broader inventive principles disclosed. As may be easily understood from the foregoing, the basic concepts of the present inventive technology may be embodied in a variety of ways. It generally involves systems, methods, techniques as well as devices to accomplish a variable geometry manifold configured to generate a laminar air flow. In this application, the methods and apparatus for the aforementioned systems are disclosed as part of the results shown to be achieved by the various devices described and as steps which are inherent to utilization. They are simply the natural result of utilizing the devices as intended and described. In addition, while some devices are disclosed, it should be understood that these not only accomplish certain methods but also can be varied in a number of

ways. Importantly, as to all of the foregoing, all of these facets should be understood to be encompassed by this disclosure.

[0051] While the invention has been described in connection with a preferred embodiment, it is not intended to limit the scope of the invention to the particular form set forth, but on the contrary, it is intended to cover such alternatives, modifications, and equivalents as may be included within the spirit and scope of the invention as defined by the statements of invention. As can be easily understood from the foregoing, the basic concepts of the present invention may be embodied in a variety of ways. It involves both techniques as well as devices to accomplish the appropriate system. In this application, the techniques are disclosed as part of the results shown to be achieved by the various devices described and as steps which are inherent to utilization. They are simply the natural result of utilizing the devices as intended and described. In addition, while some devices are disclosed, it should be understood that these not only accomplish certain methods but also can be varied in a number of ways. Importantly, as to all of the foregoing, all of these facets should be understood to be encompassed by this disclosure.

[0052] The discussion included in this application is intended to serve as a basic description. The reader should be aware that the specific discussion may not explicitly describe all embodiments possible; many alternatives are implicit. It also may not fully explain the generic nature of the invention and may not explicitly show how each feature or element can actually be representative of a broader function or of a great variety of alternative or equivalent elements. Again, these are implicitly included in this disclosure. Where the invention is described in method-oriented terminology, each element of the claims corresponds to a device. Apparatus claims may not only be included for the device described, but also method or process claims may be included to address the functions the invention and each element performs. Neither the description nor the terminology is intended to limit the scope of the claims that will be included in any subsequent patent application.

[0053] It should also be understood that a variety of changes may be made without departing from the essence of the invention. Such changes are also implicitly included in the description. They still fall within the scope of this invention. A broad disclosure encompassing both the explicit embodiment(s) shown, the great variety of implicit alternative embodiments, and the broad methods or processes and the like are encompassed by this disclosure and may be relied upon when drafting any claims. It should be understood that such language changes and broader or more detailed claiming may be accomplished at a later date (such as by any required deadline) or in the event the applicant subsequently seeks a patent filing based on this filing. With this understanding, the reader should be aware that this disclosure is to be understood to support any subsequently filed patent application that may seek examination of as broad a base of claims as deemed within the applicant's right and may be designed to yield a patent covering numerous aspects of the invention both independently and as an overall system.

[0054] Further, each of the various elements of the invention and claims may also be achieved in a variety of manners. Additionally, when used or implied, an element is to be understood as encompassing individual as well as plural structures that may or may not be physically connected. This disclosure should be understood to encompass each such variation, be it a variation of an embodiment of any apparatus embodiment, a method or process embodiment, or even merely a variation of any element of these. Particularly, it should be understood that as the disclosure relates to elements of the invention, the words for each element may be expressed by equivalent apparatus terms or method terms—even if only the function or result is the same. Such equivalent, broader, or even more generic terms should be considered to be encompassed in the description of each element or action. Such terms can be substituted where desired to make explicit the implicitly broad coverage to which this invention is entitled. As but one example, it should be understood that all actions may be expressed as a means for taking that action or as an element which causes that action. Similarly, each physical element disclosed should be understood to encompass a disclosure

of the action which that physical element facilitates. Regarding this last aspect, as but one example, the disclosure of a “coupler” should be understood to encompass disclosure of the act of “coupling”—whether explicitly discussed or not—and, conversely, were there effectively disclosure of the act of “coupling”, such a disclosure should be understood to encompass disclosure of a “coupling method and/or technique, and/or device.” Such changes and alternative terms are to be understood to be explicitly included in the description.

[0055] Any patents, publications, or other references mentioned in this application for patent, such as in the specification or an IDS are hereby incorporated herein by reference in their entirety. Any priority case(s) claimed by this application is hereby appended and hereby incorporated herein by reference in their entirety. In addition, as to each term used it should be understood that unless its utilization in this application is inconsistent with a broadly supporting interpretation, common dictionary definitions should be understood as incorporated for each term and all definitions, alternative terms, and synonyms such as contained in the Random House Webster's Unabridged Dictionary, second edition are hereby incorporated herein by reference in their entirety. Finally, all references listed in the list of References To Be Incorporated By Reference In Accordance With The Patent Application or other information disclosure statement and the like filed with the application are hereby appended and hereby incorporated herein by reference in their entirety, however, as to each of the above, to the extent that such information or statements incorporated by reference might be considered inconsistent with the patenting of this/these invention(s) such statements are expressly not to be considered as made by the applicant(s).

[0056] Thus, the applicant(s) should be understood to have support to claim and make a statement of invention to at least: i) each of the methods and/or apparatus for providing a variable geometry manifold for generating laminar air flow through an electrostatic precipitation device as herein disclosed and described, ii) the related methods disclosed and described, iii) similar, equivalent, and even implicit variations of each of these devices and methods, iv) those alternative designs which accomplish each of the functions shown as are disclosed and described, v) those alternative designs and methods which accomplish each of the functions shown as are implicit to accomplish that which is disclosed and described, vi) each feature, component, and step shown as separate and independent inventions, vii) the applications enhanced by the various systems or components disclosed, viii) the resulting products produced by such systems or components, ix) each system, method, and element shown or described as now applied to any specific field or devices mentioned, x) methods and apparatuses substantially as described hereinbefore and with reference to any of the accompanying examples, xi) the various combinations and permutations of each of the elements disclosed, xii) each potentially dependent claim or concept as a dependency on each and every one of the independent claims or concepts presented, and xiii) all inventions described herein.

[0057] With regard to claims whether now or later presented for examination, it should be understood that for practical reasons and so as to avoid great expansion of the examination burden, the applicant may at any time present only initial claims or perhaps only initial claims with only initial dependencies. The office and any third persons interested in potential scope of this, or subsequent applications should understand that broader claims may be presented at a later date in this case, in a case claiming the benefit of this case, or in any continuation in spite of any preliminary amendments, other amendments, claim language, or arguments presented, thus throughout the pendency of any case there is no intention to disclaim or surrender any potential subject matter. It should be understood that if or when broader claims are presented, such may require that any relevant prior art that may have been considered at any prior time may need to be re-visited since it is possible that to the extent any amendments, claim language, or arguments presented in this or any subsequent application are considered as made to avoid such prior art, such reasons may be eliminated by later presented claims or the like. Both the examiner and any person otherwise interested in existing or later potential coverage, or considering if there has at any time been any possibility of an indication of disclaimer or surrender of potential coverage, should be

aware that no such surrender or disclaimer is ever intended or ever exists in this or any subsequent application. Limitations such as arose in *Hakim v. Cannon Avent Group, PLC*, 479 F.3d 1313 (Fed. Cir 2007), or the like are expressly not intended in this or any subsequent related matter. In addition, support should be understood to exist to the degree required under new matter laws—including but not limited to European Patent Convention Article 123(2) and United States Patent Law 35 USC 132 or other such laws—to permit the addition of any of the various dependencies or other elements presented under one independent claim or concept as dependencies or elements under any other independent claim or concept. In drafting any claims at any time whether in this application or in any subsequent application, it should also be understood that the applicant has intended to capture as full and broad a scope of coverage as legally available. To the extent that insubstantial substitutes are made, to the extent that the applicant did not in fact draft any claim so as to literally encompass any particular embodiment, and to the extent otherwise applicable, the applicant should not be understood to have in any way intended to or actually relinquished such coverage as the applicant simply may not have been able to anticipate all eventualities; one skilled in the art, should not be reasonably expected to have drafted a claim that would have literally encompassed such alternative embodiments.

[0058] Further, if or when used, the use of the transitional phrase “comprising” is used to maintain the “open-end” claims herein, according to traditional claim interpretation. Thus, unless the context requires otherwise, it should be understood that the term “comprise” or variations such as “comprises” or “comprising”, are intended to imply the inclusion of a stated element or step or group of elements or steps but not the exclusion of any other element or step or group of elements or steps. Such terms should be interpreted in their most expansive form so as to afford the applicant the broadest coverage legally permissible. It should be understood that this application also provides support for any combination of elements in the claims and even incorporates any desired proper antecedent basis for certain claim combinations such as with combinations of method, apparatus, process, and the like claims.

[0059] Any claims set forth at any time are hereby incorporated by reference as part of this description of the invention, and the applicant expressly reserves the right to use all of or a portion of such incorporated content of such claims as additional description to support any of or all of the claims or any element or component thereof, and the applicant further expressly reserves the right to move any portion of or all of the incorporated content of such claims or any element or component thereof from the description into the claims or vice-versa as necessary to define the matter for which protection is sought by this application or by any subsequent continuation, division, or continuation-in-part application thereof, or to obtain any benefit of, reduction in fees pursuant to, or to comply with the patent laws, rules, or regulations of any country or treaty, and such content incorporated by reference shall survive during the entire pendency of this application including any subsequent continuation, division, or continuation-in-part application thereof or any reissue or extension thereon. The inventive subject matter is to include, but certainly not be limited as, a system substantially as herein described with reference to any one or more of the Figures and Description (including the following: for example, the process according to any claims and further comprising any of the steps as shown in any Figures, separately, in any combination or permutation).

Claims

1. A system for electrostatically purifying air using a variable geometry manifold comprising: an axially symmetric intake having a proximal and terminal end, wherein the terminal end is narrower than the proximal end, and wherein the axially symmetric intake is configured to be in fluid communication with an electrostatic precipitator (ESP) air purifier; a suction fan mounted to the terminal end of the axially symmetric intake, and configured to draw external air containing

particulate matter that is neutrally charged through the axially symmetric intake, wherein the axially symmetric intake forms a laminar flow of the external air that is introduced to the ESP air purifier where it passes through an electric field that extends from a negatively charged plate to a positively charged collecting plate; and wherein the electric field ionizes the particulate matter within the laminal flow, and wherein the electric field repels the ionized particulate matter away from the negative plate and towards the collecting plate causing the ionized particulate matter to attach to the collecting plate.

2. The system of claim 1, wherein said axially symmetric intake comprises a continuous axially symmetric intake, or a non-continuous axially symmetric intake.

3. (canceled)

4. The system of claim 2, wherein said continuous axially symmetric intake comprises a plurality of intake panels coupled by one or more in-line brackets.

5. The system of claim 2, wherein said continuous axially symmetric intake comprises at least four intake panels coupled by a plurality in-line brackets.

6. The system of claim 5, wherein said plurality of in-line brackets externally couple the lateral surfaces of the four intake panels forming a joint that maintains a continuous inner surface of the axially symmetric intake.

7. The system of claim 1, further comprising an intake assembly coupled to the proximal end of the axially symmetric intake.

8-11. (canceled)

12. The system of claim 1, further comprising a support base securing the axially symmetric intake in-line with the air flow into the body of the ESP air purifier.

13. The system of claim 1, wherein said support base comprises a plurality of support arms secured to the proximal end of the axially symmetric intake.

14. The system of claim 1, wherein said support base comprises one or more support columns secured to the terminal end of the axially symmetric intake.

15. The system of claim 1, further comprising a mounting plate configured to secure the suction fan in-line with the air flow into the body of the ESP air purifier.

16-37. (canceled)

38. A system for electrostatically purifying air using a variable geometry manifold comprising: an axially symmetric intake having a proximal and terminal end, wherein the terminal end is narrower than the proximal end, and wherein the axially symmetric intake is configured to be in fluid communication with an electrostatic precipitator (ESP) air purifier; wherein the manifold is secured to a stationary or non-stationary object and positioned to allow external air containing particulate matter that is neutrally charged to pass through the axially symmetric intake, wherein the axially symmetric intake forms a laminar flow of the external air that is introduced to the ESP air purifier where it passes through an electric field that extends from a negatively charged plate to a positively charged collecting plate; and wherein the electric field ionizes the particulate matter within the laminal flow, and wherein the electric field repels the ionized particulate matter away from the negative plate and towards the collecting plate causing the ionized particulate matter to attach to the collecting plate.

39. The system of claim 38, wherein said axially symmetric intake comprises a continuous axially symmetric intake, or a non-continuous, axially symmetric intake.

40. The system of claim 38, wherein said non-stationary object is selected from: a vehicle, a bus, a car, a train, a boat, a ship, an airplane.

41. The system of claim 39, wherein said continuous axially symmetric intake comprises a plurality of intake panels coupled by one or more in-line brackets.

42. The system of claim 39, wherein said continuous axially symmetric intake comprises at least four intake panels coupled by a plurality in-line brackets.

43. The system of claim 42, wherein said plurality of in-line brackets externally couple the lateral

surfaces of the four intake panels forming a joint that maintains a continuous inner surface of the axially symmetric intake.

44. The system of claim 42, further comprising an intake assembly coupled to the proximal end of the axially symmetric intake.

45-48. (canceled)

49. The system of claim **48**, further comprising a support base securing the axially symmetric intake in-line with the air flow into the body of the ESP air purifier.

50. The system of claim 38, wherein said support base comprises a plurality of support arms secured to the proximal end of the axially symmetric intake.

51. The system of claim 38, wherein said support base comprises one or more support columns secured to the terminal end of the axially symmetric intake.

52-53. (canceled)
